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Roll No

EE-8003(2) (CBGS)

B.E. VIII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Electrical Hybrid Vehicles

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What is the need and importance of electric and hybrid electric vehicles?
b) With the help of a neat block diagram explain different subsystems of electric drive train.
2. Describe in detail all modes of recent operation for series hybrid vehicle.
3. Draw and explain architecture and power flow control of series and parallel hybrid electric drive train.
4. a) Explain the two quadrant operation of chopper DC motor drive with suitable waveform for electrical vehicle.
b) Analyze the performance of induction motors for electric and hybrid electric vehicle application.
5. Explain the fuel cell and flywheel as energy source elements in electric and Hybrid electric vehicles.

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6. Describe the battery charging methods used EV/HEV in detail.
7. Classify and explain the basic principle of rule based energy management system.
8. Explain the following (any two)
 - i) Super capacitors as energy source elements in electric and Hybrid electric vehicles.
 - ii) Design of a Battery Electric Vehicle (BEV).
 - iii) Comparison of different energy management strategies.
 - iv) Hybridization of different energy storage devices.
